

**Assignment 1**  
**30 Marks**  
**Deadline: Nov 1, 2025 (Saturday)**

**Question - 1:**

- a. **Explain** Amdahl's Law in your own words. [2]
- b. Can you relate/connect Amdahl's law with any of the design principles, mentioned below? Explain the reason with an example. [3]

|   | Design Principle                   |
|---|------------------------------------|
| 1 | Performance via Prediction         |
| 2 | Performance via Pipelining         |
| 3 | Make the common case faster        |
| 4 | Use Abstraction to Simplify Design |

**Question - 2:**

Define the term throughput and response time. How do these two differ, and why might one be prioritized over the other in different computing environments?  
[1+1+2]

**Question - 3:**

Consider three different processors P1, P2, and P3 executing the exact same instruction set. P1 has a 4 GHz clock rate and a CPI of 2. P2 has a 2.5 GHz clock rate and a CPI of 1.5. P3 has a 5.0 GHz clock rate and has a CPI of 2.7.

Now answer the following questions:

- a. Which processor has the highest performance expressed in instructions per second?

- b. We are trying to reduce the execution time by 40%, but this leads to an increase of 35% in the CPI. What clock rate should we have to get this time reduction?  
[2+1]

#### **Question - 4:**

What do you understand by “**Performance Via Prediction**” in terms of computer architecture? Give a proper example of it. [2+1]

#### **Question - 5:**

Describe how abstraction helps in simplifying the design of complex computer systems. Give a layered example showing abstraction from hardware to software.  
[2 + 1]

#### **Question - 6:**

Two computers, A and B, execute the same set of instructions.

- Computer A: Clock Rate = 3 GHz, CPI = 1.5
- Computer B: Clock Rate = 2.5 GHz, CPI = 1.2

Which one is faster? Show calculations for CPU time. [1 + 1 + 1]

#### **Question - 7:**

A processor has the following instruction mix:

- 50% ALU instructions with CPI = 1
- 30% load/store with CPI = 2
- 20% branch with CPI = 3

Calculate the **average CPI** of the program. [3]

### Question - 8:

A computer runs a program in 20 seconds. A compiler optimization reduces instruction count by 15% and CPI by 10%, but the clock rate stays the same. What is the new execution time? [3]

### Question - 9:

To calculate the **benchmark** of a system, why do we take the **geometric mean** instead of only taking the **average** of the individual spec ratios? [3]